

1. Evaluate $V_{AB} = V_A - V_B$ from the circuit shown in Figure 1.

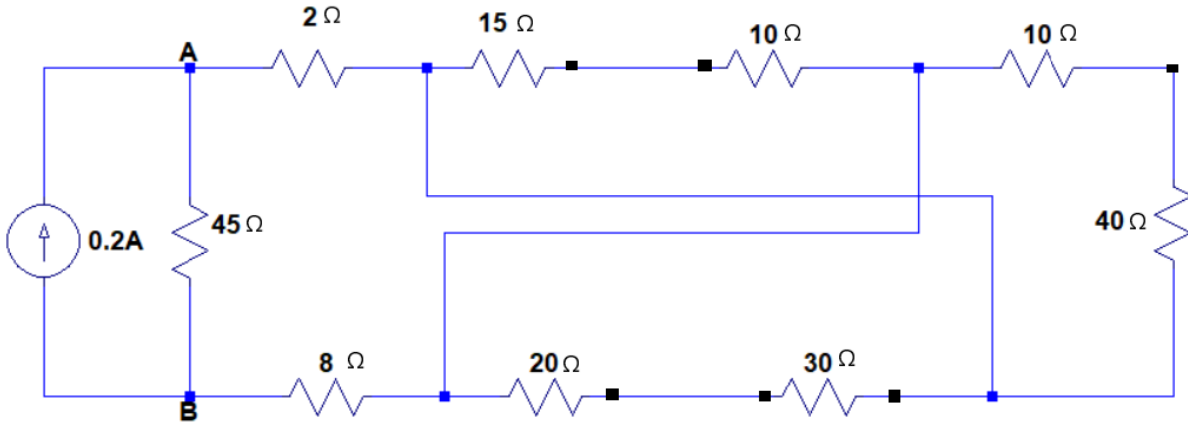


Figure 1: Evaluate V_{AB} from the above circuit.

2. Apply Ohm's Law, KCL and KVL to evaluate I_s , v_x and I_{in} from the circuit shown in Figure 2.

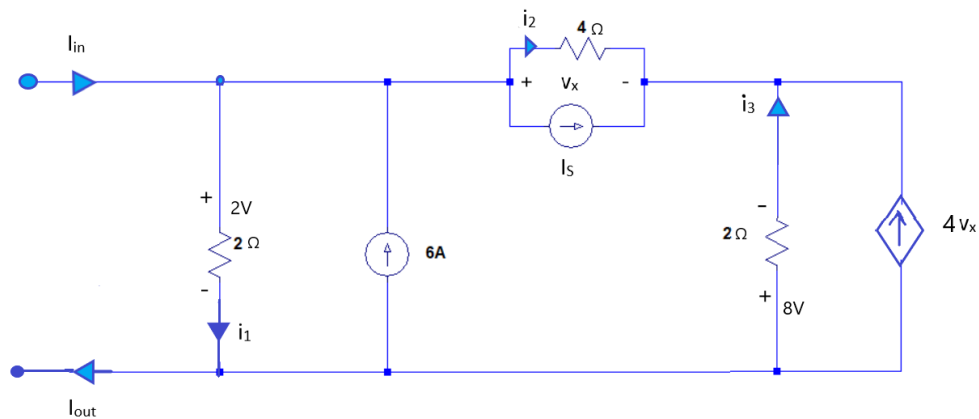


Figure 2: Evaluate I_s , v_x and I_{in} from the above circuit.

3. Obtain the truth table for the circuits shown below in Figure 3. Are they recognizable as any of the standard logic gates?

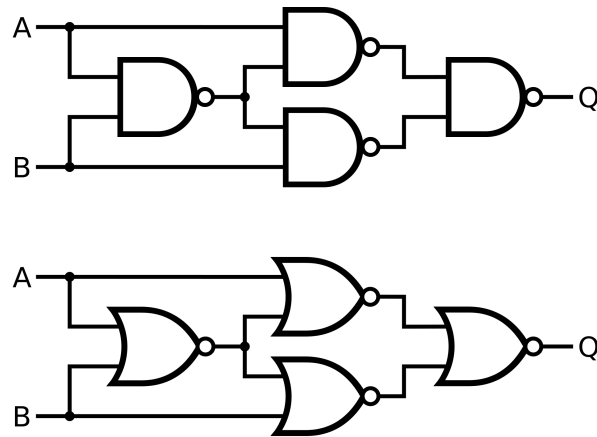


Figure 3: Examples of NAND logic and NOR logic

4. Consider the circuit shown in the figure below.
- Obtain the truth table for the output F in terms of the inputs A , B and C .
 - Obtain an algebraic expression for the output F in terms of the inputs A , B and C .

